

NETWORK SECURITY ASSESSMENT

(Using KALI LINUX and Nmap)

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➤ ABSTRACT

This project focuses on performing a network security assessment of an authorized system using Kali Linux and Nmap to identify ports, running services and possible security risks.

➤ OBJECTIVE

- Identify active network services
- Find open ports
- Analyze exposed services
- Understand possible security weaknesses
- Suggest security improvements

➤ TOOLS AND TECHNOLOGIES USED

<u>TOOLS</u>	<u>PURPOSE</u>
Kali Linux	Security testing environment
Ping	Checking activity
Nmap	Network scanning
WHOIS	Information gathering
Linux Terminal	Command execution

➤ LAB SETUP

Testing Machine: Kali Linux

Target: Authorized personal system

Network Type: Local Network

Target IP: 10.137.xxx.xxx (my private IP)

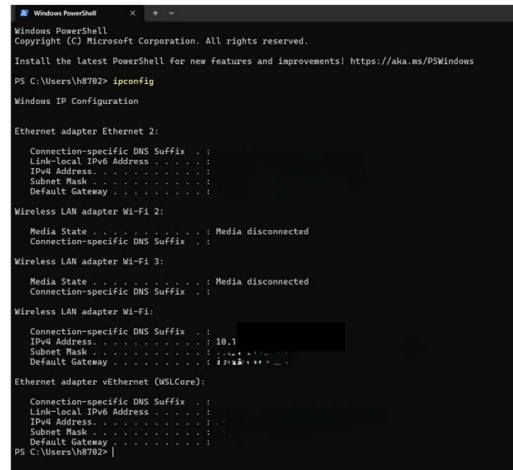
➤ METHODOLOGY

STEP 1: IP IDENTIFICATION

COMMAND : `ipconfig`

PURPOSE : The target IP address was identified from the network configuration.

WORK :



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\h8702> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::...
    IPv4 Address. . . . . : 
    Subnet Mask . . . . . : 
    Default Gateway . . . . . : 

Wireless LAN adapter Wi-Fi 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi 3:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    IPv4 Address. . . . . : 10.1...
    Subnet Mask . . . . . : 
    Default Gateway . . . . . : 

Ethernet adapter vEthernet (WSLCore):

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::...
    IPv4 Address. . . . . : 
    Subnet Mask . . . . . : 
    Default Gateway . . . . . : 

PS C:\Users\h8702>
```

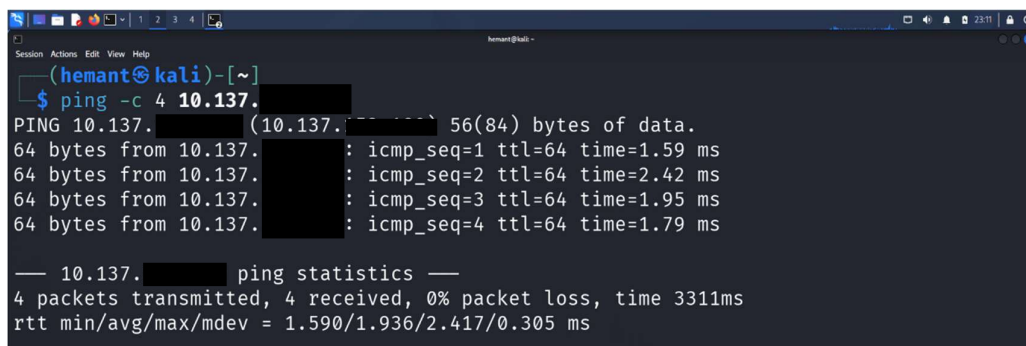
EXPLANATION : The IP stands for “Internet protocols” which is an address for a system or group of systems which possess the information about that system.

STEP 2: PING SCAN

COMMAND : `ping -c 4 10.137.xxx.xxx`

PURPOSE : To check whether the target is online or offline

WORK :



```
hemant@kali: ~
$ ping -c 4 10.137.
PING 10.137. (10.137. 56(84) bytes of data:
64 bytes from 10.137. : icmp_seq=1 ttl=64 time=1.59 ms
64 bytes from 10.137. : icmp_seq=2 ttl=64 time=2.42 ms
64 bytes from 10.137. : icmp_seq=3 ttl=64 time=1.95 ms
64 bytes from 10.137. : icmp_seq=4 ttl=64 time=1.79 ms

— 10.137. ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3311ms
rtt min/avg/max/mdev = 1.590/1.936/2.417/0.305 ms
```

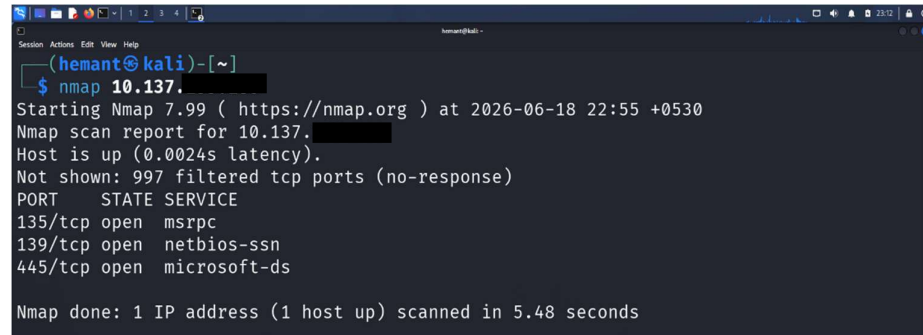
EXPLANATION : Here it says , that total 4 packages were sent to the host and in return it received 4 packages which strongly justifies that the system is online to which it was requested .

STEP 3: NETWORK SCAN

COMMAND : Nmap<IP>

PURPOSE : To identify open and closed ports.

WORK :



```
(hemant@kali)-[~]
$ nmap 10.137.1.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-18 22:55 +0530
Nmap scan report for 10.137.1.1
Host is up (0.0024s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 5.48 seconds
```

EXPLANATION : Nmap is also known as Network Mapper which is a network scanning tool used to discover hosts ,identify open ports ,detect running services and information about systems on a network .This is also one of a fundamental step to find vulnerabilities in the system which can be later secured from network spoofing and damages.

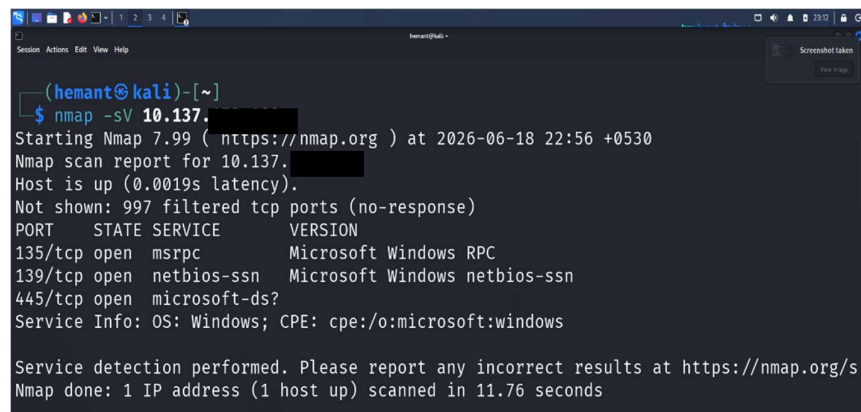
Here the report is showing host is active, ports are also open on tcp here and the services are running too in three handshakes mode.

STEP 4:SERVICE DETECTION

COMMAND: nmap -sV < Target IP>

PURPOSE :To identify services running on open ports and to gather version information.

WORK :



```
(hemant@kali)-[~]
$ nmap -sV 10.137.1.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-18 22:56 +0530
Nmap scan report for 10.137.1.1
Host is up (0.0019s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/s
Nmap done: 1 IP address (1 host up) scanned in 11.76 seconds
```

EXPLANATION: The nmap -sV command was used to perform service version detection on the target system. The scan confirmed that the host was active and identified several open ports, including 135 (MSRPC), 139 (NetBIOS Session service), and 445 (Microsoft Directory Service). It also detected that the target system was running on MS Windows. This information helps in understanding the service exposed on the network and assessing potential security risk.

Saving the nmap scanning :

```
(hemant@kali)-[~]
$ nmap 10.137. -oN nmap-assessment.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-18 23:02 +0530
Nmap scan report for 10.137.
Host is up (0.0013s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 5.09 seconds
```

Opening the nmap file :

```
(hemant@kali)-[~]
$ cat nmap-assessment.txt
# Nmap 7.99 scan initiated Thu Jun 18 23:02:17 2026 as: /usr/lib/nmap/nmap --privilege
session.txt 10.137.
Nmap scan report for 10.137.
Host is up (0.0013s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds

# Nmap done at Thu Jun 18 23:02:22 2026 -- 1 IP address (1 host up) scanned in 5.09 se
```

➤ ADDITIONAL RECONNAISSANCE

COMMAND: whois <target IP>

PURPOSE : To get more information about the host and server.

WORK : (refer to next page)

```
hemant@kali:~$ whois 47.31
#
# ARIN WHOIS data and services are subject to the Terms of Use
# available at: https://www.arin.net/resources/registry/whois/tou/
#
# If you see inaccuracies in the results, please report at
# https://www.arin.net/resources/registry/whois/inaccuracy\_reporting/
#
# Copyright 1997-2026, American Registry for Internet Numbers, Ltd.
#

NetRange:      47.29.0.0 - 47.31.255.255
CIDR:          47.30.0.0/15, 47.29.0.0/16
NetName:       APNIC
NetHandle:     NET-47-29-0-0-1
Parent:        NET47 (NET-47-0-0-0-0)
```

```
hemant@kali:~$ whois 47.31
CIDR:          47.30.0.0/15, 47.29.0.0/16
NetName:       APNIC
NetHandle:     NET-47-29-0-0-1
Parent:        NET47 (NET-47-0-0-0-0)
NetType:       Early Registrations, Transferred to APNIC
OriginAS:
Organization:  Asia Pacific Network Information Centre (APNIC)
RegDate:       2016-02-09
Updated:       2016-02-09
Ref:           https://rdap.arin.net/registry/ip/47.29.0.0

ResourceLink:  https://apps.db.ripe.net/db-web-ui/query
ResourceLink:  whois://whois.apnic.net

OrgName:       Asia Pacific Network Information Centre
OrgId:         APNIC
Address:       PO Box 3646
City:          South Brisbane
```

```
hemant@kali:~$ whois 47.31
City:          South Brisbane
StateProv:     QLD
PostalCode:    4101
Country:       AU
RegDate:
Updated:       2012-01-24
Ref:           https://rdap.arin.net/registry/entity/APNIC

ReferralServer: whois://whois.apnic.net
ResourceLink:   http://wq.apnic.net/whois-search/static/search.html

OrgAbuseHandle: AWC12-ARIN
OrgAbuseName:   APNIC Whois Contact
OrgAbusePhone:  +61 7 3858 3188
OrgAbuseEmail:  search-apnic-not-arin@apnic.net
OrgAbuseRef:    https://rdap.arin.net/registry/entity/AWC12-ARIN

OrgTechHandle: AWC12-ARIN
OrgTechName:   APNIC Whois Contact
```

```
Session Actions Edit View Help
OrgTechName:  APNIC Whois Contact
OrgTechPhone: +61 7 3858 3188
OrgTechEmail: search-apnic-not-arin@apnic.net
OrgTechRef:   https://rdap.arin.net/registry/entity/AWC12-ARIN

#
# ARIN WHOIS data and services are subject to the Terms of Use
# available at: https://www.arin.net/resources/registry/whois/tou/
#
# If you see inaccuracies in the results, please report at
# https://www.arin.net/resources/registry/whois/inaccuracy_reporting/
#
# Copyright 1997-2026, American Registry for Internet Numbers, Ltd.
#

Found a referral to whois.apnic.net.
```

```
Session Actions Edit View Help
Found a referral to whois.apnic.net.

% [whois.apnic.net]
% Whois data copyright terms   http://www.apnic.net/db/dbcopyright.html

% Information related to '47.31.0.0 - 47.31.255.255'

% Abuse contact for '47.31.0.0 - 47.31.255.255' is 'ip.abuse@ril.com'

inetnum:        47.31.0.0 - 47.31.255.255
netname:        RELIANCEJIO-IN
descr:          Reliance Jio infocomm ltd
country:        IN
admin-c:        RJIL1-AP
tech-c:         RJIL1-AP
abuse-c:        AI808-AP
status:         ALLOCATED NON-PORTABLE
mnt-by:         MAINT-IN-RELIANCEJIO
mnt-irt:        IRT-INFOTELR4G-IN
```

```
Session Actions Edit View Help

country:        IN
admin-c:        RJIL1-AP
tech-c:         RJIL1-AP
abuse-c:        AI808-AP
status:         ALLOCATED NON-PORTABLE
mnt-by:         MAINT-IN-RELIANCEJIO
mnt-irt:        IRT-INFOTELR4G-IN
last-modified:  2021-02-04T11:06:15Z
source:         APNIC

irt:            IRT-INFOTELR4G-IN
address:        Reliance Jio Infocomm Ltd RCP Ghansoli
e-mail:         ip.abuse@ril.com
abuse-mailbox:  ip.abuse@ril.com
admin-c:        IBSP1-AP
tech-c:         IBSP1-AP
auth:           # Filtered
remarks:        ip.abuse@ril.com was validated on 2026-02-18
mnt-by:         MAINT-RELIANCE-INFOTEL-IN
```

```
Session Actions Edit View Help
mnt-by: MAINT-RELIANCE-INFOTEL-IN
last-modified: 2026-02-18T13:42:50Z
source: APNIC

role: ABUSE INFOTELR4GIN
country: ZZ
address: Reliance Jio Infocomm Ltd RCP Ghansoli
phone: +000000000
e-mail: ip.abuse@ril.com
admin-c: IBSP1-AP
tech-c: IBSP1-AP
nic-hdl: AI808-AP
remarks: Generated from irt object IRT-INFOTELR4G-IN
remarks: ip.abuse@ril.com was validated on 2026-02-18
abuse-mailbox: ip.abuse@ril.com
mnt-by: APNIC-ABUSE
last-modified: 2026-02-18T13:43:20Z
source: APNIC
```

```
Session Actions Edit View Help
nic-hdl: RJIL1-AP
mnt-by: MAINT-IN-RELIANCEJIO
last-modified: 2016-03-09T23:55:07Z
source: APNIC

% Information related to '47.31.64.0/21AS55836'

route: 47.31.64.0/21
origin: AS55836
descr: Reliance Jio Infocomm Limited
Reliance Coporate IT park LTD
Ghansoli NaviMumbai
mnt-by: MAINT-IN-RELIANCEJIO
last-modified: 2025-06-24T07:59:03Z
source: APNIC

% This query was served by the APNIC Whois Service version 1.88.48 (WHOIS-JP1)
```

➤ FINDINGS

Port	Service	Status
22	SSH	INACTIVE
135	TCP	ACTIVE
139	TCP	ACTIVE
445	TCP	ACTIVE
80	HTTPS	INACTIVE

❖ SECURITY ANALYSIS

- **SSH**

Purpose : Remote administration

Risk: Weak authentication can increase security risk.

Recommendation : Use strong passwords and secure configurations .

- **HTTP**

Purpose : Web service

Risk: Communication is not encrypted.

Recommendations : Use HTTPS.

- **TCP**

Purpose: It provides reliable communication between devices.

Risk : Open TCP ports can increase attack surface.

Recommendation : Keep only required ports open and secure exposed services.

➤ CONCLUSION

The assessment successfully identified exposed network services and demonstrated the importance of reducing unnecessary open ports and maintaining secure configurations.